



AI SOFTWARE DEVELOPMENT INTERNSHIP

EMAN MASOOD (BSSE)

Week 6 — Detailed Task Sheet

LangChain & LCEL • Advanced Retrieval • Tool Calling Resilient .NET fi AI
Service Integration • End-to-End Streaming • Git Rebase

PART C — STRUCTURED OUTPUT & CONVERSATION MEMORY

Outputs

```
• (venv) PS D:\Week6\Week6_PartC_Structured_Memory> python structured_output.py
D:\Week6\Week6_PartC_Structured_Memory\venv\Lib\site-packages\langchain_google_genai\chat_models.py:3908: UserWarning: Model 'gemini-3.5-flash-lite' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
Direct use of automatic function calling (AFC) in Models.generate_content is not recommended. Instead, we recommend to use AFC in Chat.send_message. Similarly, direct use of AFC in Models.generate_content_stream is not recommended. Instead, we recommend to use AFC in Chat.send_message_stream.
=====
PART C - STRUCTURED OUTPUT
=====

QUESTION:
Which books are science fiction?

STRUCTURED RESULT:
Answer: Dune by Frank Herbert is a science fiction novel.
Confidence: high
Sources: ['Dune']

Result type: BookAnswer

Structured output completed successfully.
• (venv) PS D:\Week6\Week6_PartC_Structured_Memory>
```

```
(venv) PS D:\Week6\Week6_PartC_Structured_Memory> python conversation_memory.py
D:\Week6\Week6_PartC_Structured_Memory\conversation_memory.py:6: DeprecationWarning: 'langchain-community' is being sunset and is no longer actively maintained. See https://github.com/langchain-ai/langchain-community/issues/674 for details and migration guidance toward standalone integration packages.
  from langchain_community.chat_message_histories import ChatMessageHistory
DeprecationWarning: RunnableChatMessageHistory is deprecated. Use LangGraph's built-in persistence instead.
=====
PART C - CONVERSATION MEMORY
=====

SESSION: user-42
D:\Week6\Week6_PartC_Structured_Memory\conversation_memory.py:41: LangChainDeprecationWarning: The class 'InMemoryChatMessageHistory' was deprecated in LangChain 1.6.4 and will be removed in 2.8.0. See the short-term memory documentation for recommended alternatives: https://docs.langchain.com/oss/python/langchain/short-term-memory
  session_store[session_id] = ChatMessageHistory()
Direct use of automatic function calling (AFC) in Models.generate_content is not recommended. Instead, we recommend to use AFC in Chat.send_message. Similarly, direct use of AFC in Models.generate_content_stream is not recommended. Instead, we recommend to use AFC in Chat.send_message_stream.

Question 1:
Tell me about Dune by Frank Herbert.

Answer 1:
[{'type': 'text', 'text': 'I do not have enough context about "Dune" by Frank Herbert in our conversation history to tell you about the book.', 'extras': {'signature': 'f6a0d6f9f8f8c80x545c0h1/1o1ck3p10uhyf8UkA512h0DQcUP86TaspI27ng8QvU4h18VCF2uW58Wx02h2swWfouk2qkDFQqqr8H7F7F2mAM02i67zy8E1-'}}]

Question 2 (follow-up):
What genre is it?

Answer 2:
[{'type': 'text', 'text': 'Based on the conversation history, "Dune" by Frank Herbert is a science fiction novel.', 'extras': {'signature': 'f6a0d6f9f8f8c80x545c0h1/1o1ck3p10uhyf8UkA512h0DQcUP86TaspI27ng8QvU4h18VCF2uW58Wx02h2swWfouk2qkDFQqqr8H7F7F2mAM02i67zy8E1-'}}]

=====
SESSION: user-99 (New Session)
=====

Question:
What genre is it?

Answer from new session:
[{'type': 'text', 'text': 'I do not have enough context to answer this question.', 'extras': {'signature': 'f6a0d6f9f8f8c80x545c0h1/1o1ck3p10uhyf8UkA512h0DQcUP86TaspI27ng8QvU4h18VCF2uW58Wx02h2swWfouk2qkDFQqqr8H7F7F2mAM02i67zy8E1-'}}]

Conversation memory test completed successfully.
(venv) PS D:\Week6\Week6_PartC_Structured_Memory>
```

```

(venv) PS D:\Week6\Week6_Part_Structured_Memory> python structured_memory_chain.py
D:\Week6\Week6_Part_Structured_Memory>structured_memory_chain.py:3: DeprecationWarning: 'langchain-community' is being sunset and is no longer actively maintained. See https://github.com/langchain-ai/langchain-community/issues/674 for details and migration guidance toward
standalone integration packages.
  from langchain_community.chat_message_histories import ChatMessageHistory
<sys:0: LangChainDeprecationWarning: RunnableWithMessageHistory is deprecated. Use LangGraph's built-in persistence instead.
=====
PART C - STRUCTURED OUTPUT + CONVERSATION MEMORY
=====

SESSION: user-42
D:\Week6\Week6_Part_Structured_Memory>structured_memory_chain.py:37: LangChainDeprecationWarning: The class 'InMemoryChatMessageHistory' was deprecated in LangChain 1.6.4 and will be removed in 2.0.0. See the short-term memory documentation for recommended alternatives: ht
tps://docs.langchain.com/docs/python/langchain/short-term-memory
  session_store[session_id] = ChatMessageHistory()
Direct use of Automatic Function Calling (AFC) in Models.generate_content is not recommended. Instead, we recommend to use AFC in Chat.send_message. Similarly, direct use of AFC in Models.generate_content_stream is not recommended. Instead, we recommend to use AFC in Chat.
send_message_stream.

Question 1:
Dune by Frank Herbert is a science fiction novel. Tell me about this book.

Structured Answer 1:
Answer: There is not enough context in the conversation history to provide details about Dune by Frank Herbert.
Confidence: High
Sources: []

Result type: BookAnswer

Question 2 (follow-up):
What genre is it?

Structured Answer 2:
Answer: Dune by Frank Herbert is a science fiction novel.
Confidence: High
Sources: ['Dune']

Result type: BookAnswer

=====
SESSION: user-99 (New Session)
=====

Question:
What genre is it?

Structured Answer from New Session:
Answer: There is not enough context in the conversation history to determine what book or "it" refers to, so the genre cannot be identified.
Confidence: High
Sources: []

Result type: BookAnswer

=====
Structured output + memory test completed successfully.
=====
(venv) PS D:\Week6\Week6_Part_Structured_Memory>

```